Pokhara University
Faculty of Science and Technology

Course No.: CMP 513 (3 Credits)

Full marks: 100

Course title: **Data Structure and Algorithm Design**

Pass marks: 60

Nature of the course: Theory

Total Lectures: 45 hrs

Level: Master

Program: ME/MSc (Computer)

1. Course Description

This course is designed to provide an advanced exploration of data structures and algorithms, with a specific focus on their application in the field of computer science and computer engineering. The course delves into the efficient design, implementation, and analysis of algorithms and data structures that are critical in systems programming and high-performance computing. Students will learn to analyze the performance of algorithms, and apply advanced data structures to solve complex engineering problems.

2. General Objectives

After successful completion of this course, students should be able to:

- Design and implement advanced data structures and algorithms to achieve the need of computer applications.
- Analyze and compare the time and space complexity of algorithms and optimize them for memory, energy efficiency, and computational power.
- Solve large-scale computational problems by designing efficient data structures optimized for both software and hardware integration.

3. Methods of Instruction

Lecture, Discussion, Readings, Case Studies, Project and Research Works.

4. Contents in Detail

Specific Objectives	Contents
• Review basic data structures, algorithm design techniques and algorithm analysis tools.	**1. Introduction to Basic Data Structures and Algorithm Design (6 hrs)** 1.1. Basic Data Structures: Arrays, Stack, Queue, Linked list, Tree and Graph 1.2. Divide and Conquer, Greedy Algorithm, Backtracking Algorithms, Branch & Bound and Randomized Algorithms 1.3. Time Complexity, Space Complexity and Asymptotic notations: Big O, Big Theta and Big Omega
• Discuss and apply advanced data structures and algorithms.	**2. Advanced Data Structures and Algorithms (11 hrs)** 2.1. Advanced Trees: B and B+ Trees 2.2. Advanced Graph: Planar Graphs and its applications in Geographic and Circuit Layout Design 2.3. Network Flow Algorithms 2.3.1. Ford-Fulkerson Algorithm 2.3.2. Edmonds-Karp Algorithm 2.4. Advanced Shortest Path Algorithms 2.4.1. Bellman-Ford Algorithm 2.4.2. Johnson's algorithm 2.5. Graphical Data Structures: Quadtrees, KD Trees, R-Trees 2.6. Computational Geometry: Convex Hull and Voronoi Diagrams 2.7. Case Studies in Red-Black Tree and Its Applications, Applications of Directed Acyclic Graph, Applications of Minimum Spanning Tree in Network Design
• Familiarize with Cache Efficient Data Structures and Algorithms. • Apply the profiling tools for memory and cache efficiency analysis.	**3. Cache Efficient Data Structures (10 hrs)** 3.1. Memory Hierarchy and Cache Efficiency 3.2. Principles of Cache-Efficient Data Structures 3.3. Cache-Conscious Algorithms and Cache-Oblivious Algorithms 3.4. Profiling Tools for Analyzing Memory and Cache Efficiency 3.5. Case Studies in Cache-Conscious B Tree and Cache-Conscious Merge Sort, Cache-Oblivious merge sort, Profiling Tools: Valgrind and Cachegrind

• Discuss and apply advanced hashing techniques.	**4. Advanced Hashing Techniques (6 hrs)** 4.1. Hardware-Accelerated Hashing 4.2. Perfect Hashing 4.3. Cuckoo Hashing 4.4. Hashing in Distributed Systems 4.5. Case Studies in GPUs and FPGAs
• Discuss advanced sorting algorithms.	**5. Advanced Sorting Algorithms (6 hrs)** 5.1. Non-comparison-based sorting: Radix sort, bucket sort 5.2. Parallel Sorting Algorithms (multi-core and distributed systems) 5.3. External sorting and its applications in large-scale data engineering
• Discuss and apply the applications of dynamic programming and greedy algorithms in engineering applications.	**6. Dynamic Programming and Greedy Algorithms (6 hrs)** 6.1. Engineering applications of dynamic programming: Resource Scheduling 6.2. Greedy algorithms: Huffman coding and its relevance to data compression in embedded systems 6.3. Optimizing dynamic programming techniques for real-time systems

5. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, internal assessment, project and research works etc. The internal evaluation scheme for this course is as follows (but not limited to):

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Attendance & Class Participation	10%	60	Semester-End examination	40
Assignments	20%			
Internal Assessment	40%			
Research/Project/Case Study Work	30%			
Total Internal		60	**Total External**	40
Full Marks: 60 + 40 = 100				

Student Responsibilities

Each student must secure at least 60% marks separately in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

6. Reference Books

1. Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). *Introduction to algorithms*. MIT press.
2. Kleinberg, J. (2006). *Algorithm Design*. Addison Wesley.
3. Jain, R. (1991). *The art of computer systems performance analysis*. John Wiley & Sons.
4. Allen, W. M. (2013). *Data structures and algorithm analysis in C++*. Pearson.
5. Skiena, S. S. (2008). *The algorithm design manual* (Vol. 2). Springer.